

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z4,Z5,ZL

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z4 Z5 ZL: $\frac{Z_4Z_5Z_Lg_m-Z_4Z_L}{Z_4Z_5g_m+2Z_4Z_Lg_m+Z_4+2Z_5Z_Lg_m+2Z_L}$

$$H(z) = \frac{Z_4Z_5Z_Lg_m - Z_4Z_L}{Z_4Z_5g_m + 2Z_4Z_Lg_m + Z_4 + 2Z_5Z_Lg_m + 2Z_L}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{1}{C_5s}, \frac{1}{C_Ls} \right)$

$$H(s) = \frac{-C_5R_4s + R_4g_m}{C_5C_LR_4s^2 + 2g_m + s(2C_5R_4g_m + 2C_5 + C_LR_4g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}\sqrt{g_m}}{2C_5R_4g_m+2C_5+C_LR_4g_m}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}}$
bandwidth: $\frac{2C_5R_4g_m+2C_5+C_LR_4g_m}{C_5C_LR_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $-\frac{C_5R_4}{2C_5R_4g_m+2C_5+C_LR_4g_m}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{1}{C_5s}, \frac{R_L}{C_LR_Ls+1} \right)$

$$H(s) = \frac{-C_5R_4R_Ls + R_4R_Lg_m}{C_5C_LR_4R_Ls^2 + R_4g_m + 2R_Lg_m + s(2C_5R_4R_Lg_m + C_5R_4 + 2C_5R_L + C_LR_4R_Lg_m)}$$

Parameters:

Q: $\frac{\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}\sqrt{g_m}\sqrt{R_4+2R_L}}{2C_5R_4R_Lg_m+C_5R_4+2C_5R_L+C_LR_4R_Lg_m}$
wo: $\frac{\sqrt{R_4g_m+2R_Lg_m}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}}$
bandwidth: $\frac{\sqrt{R_4g_m+2R_Lg_m}(2C_5R_4R_Lg_m+C_5R_4+2C_5R_L+C_LR_4R_Lg_m)}{C_5C_LR_4R_L\sqrt{g_m}\sqrt{R_4+2R_L}}$
K-LP: $\frac{R_4R_L}{R_4+2R_L}$
K-HP: 0
K-BP: $-\frac{C_5R_4R_L}{2C_5R_4R_Lg_m+C_5R_4+2C_5R_L+C_LR_4R_Lg_m}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4}{C_5 C_L R_4 R_5 s^2 + 2R_4 g_m + 2R_5 g_m + s(2C_5 R_4 R_5 g_m + 2C_5 R_5 + C_L R_4 R_5 g_m + C_L R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}\sqrt{R_5}\sqrt{R_4 g_m + R_5 g_m + 1}}{2C_5 R_4 R_5 g_m + 2C_5 R_5 + C_L R_4 R_5 g_m + C_L R_4}$
wo: $\frac{\sqrt{2}R_4 g_m + 2R_5 g_m + 2}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}\sqrt{R_5}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_4 g_m + 2R_5 g_m + 2}(2C_5 R_4 R_5 g_m + 2C_5 R_5 + C_L R_4 R_5 g_m + C_L R_4)}{2C_5 C_L R_4 R_5 \sqrt{R_4 g_m + R_5 g_m + 1}}$
K-LP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$
K-HP: 0
K-BP: $-\frac{C_5 R_4 R_5}{2C_5 R_4 R_5 g_m + 2C_5 R_5 + C_L R_4 R_5 g_m + C_L R_4}$
Qz: None
Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_4 R_5 R_L s + R_4 R_5 R_L g_m - R_4 R_L}{C_5 C_L R_4 R_5 R_L s^2 + R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L + s(2C_5 R_4 R_5 R_L g_m + C_5 R_4 R_5 + 2C_5 R_5 R_L + C_L R_4 R_5 R_L g_m + C_L R_4 R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}\sqrt{R_5}\sqrt{R_L}\sqrt{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L}}{2C_5 R_4 R_5 R_L g_m + C_5 R_4 R_5 + 2C_5 R_5 R_L + C_L R_4 R_5 R_L g_m + C_L R_4 R_L}$
wo: $\frac{\sqrt{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}\sqrt{R_5}\sqrt{R_L}}$
bandwidth: $\frac{2C_5 R_4 R_5 R_L g_m + C_5 R_4 R_5 + 2C_5 R_5 R_L + C_L R_4 R_5 R_L g_m + C_L R_4 R_L}{C_5 C_L R_4 R_5 R_L}$
K-LP: $\frac{R_4 R_5 R_L g_m - R_4 R_L}{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L}$
K-HP: 0
K-BP: $-\frac{C_5 R_4 R_5 R_L}{2C_5 R_4 R_5 R_L g_m + C_5 R_4 R_5 + 2C_5 R_5 R_L + C_L R_4 R_5 R_L g_m + C_L R_4 R_L}$
Qz: None
Wz: None

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\infty, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_4 g_m + s(C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^2(C_5 C_L R_4 R_5 g_m + C_5 C_L R_4) + s(2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5 + C_L R_4 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}R_5 g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_4} \sqrt{\frac{g_m}{R_5 g_m + 1}}}{2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5 + C_L R_4 g_m}$
wo: $\sqrt{2} \sqrt{\frac{g_m}{C_5 C_L R_4 R_5 g_m + C_5 C_L R_4}}$
bandwidth: $\frac{\sqrt{2} \sqrt{\frac{g_m}{C_5 C_L R_4 R_5 g_m + C_5 C_L R_4}} (2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5 + C_L R_4 g_m)}{\sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}R_5 g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_4} \sqrt{\frac{g_m}{R_5 g_m + 1}}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_5 R_4 R_5 g_m - C_5 R_4}{2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5 + C_L R_4 g_m}$
Qz: None
Wz: None

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\infty, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_4 R_L g_m + s(C_5 R_4 R_5 R_L g_m - C_5 R_4 R_L)}{R_4 g_m + 2R_L g_m + s^2(C_5 C_L R_4 R_5 R_L g_m + C_5 C_L R_4 R_L) + s(C_5 R_4 R_5 g_m + 2C_5 R_4 R_L g_m + C_5 R_4 + 2C_5 R_5 R_L g_m + 2C_5 R_L + C_L R_4 R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_5}\sqrt{C_L}\sqrt{R_4}R_5\sqrt{R_L}g_m\sqrt{\frac{g_m}{R_5 g_m + 1}}\sqrt{R_4 + 2R_L} + \sqrt{C_5}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{g_m}{R_5 g_m + 1}}\sqrt{R_4 + 2R_L}}{C_5 R_4 R_5 g_m + 2C_5 R_4 R_L g_m + C_5 R_4 + 2C_5 R_5 R_L g_m + 2C_5 R_L + C_L R_4 R_L g_m}$

$$\begin{aligned}
\text{wo: } & \sqrt{\frac{R_4 g_m + 2 R_L g_m}{C_5 C_L R_4 R_5 R_L g_m + C_5 C_L R_4 R_L}} \\
\text{bandwidth: } & \frac{\sqrt{\frac{R_4 g_m + 2 R_L g_m}{C_5 C_L R_4 R_5 R_L g_m + C_5 C_L R_4 R_L}} (C_5 R_4 R_5 g_m + 2 C_5 R_4 R_L g_m + C_5 R_4 + 2 C_5 R_5 R_L g_m + 2 C_5 R_L + C_L R_4 R_L g_m)}{\sqrt{C_5} \sqrt{C_L} \sqrt{R_4 R_5} \sqrt{R_L g_m} \sqrt{\frac{g_m}{R_5 g_m + 1}} \sqrt{R_4 + 2 R_L} + \sqrt{C_5} \sqrt{C_L} \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{g_m}{R_5 g_m + 1}} \sqrt{R_4 + 2 R_L}} \\
\text{K-LP: } & \frac{R_4 R_L}{R_4 + 2 R_L} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_5 R_4 R_5 R_L g_m - C_5 R_4 R_L}{C_5 R_4 R_5 g_m + 2 C_5 R_4 R_L g_m + C_5 R_4 + 2 C_5 R_5 R_L g_m + 2 C_5 R_L + C_L R_4 R_L g_m} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_5 g_m + s (C_L R_5 R_L g_m - C_L R_L) - 1}{2 g_m + s^2 (2 C_4 C_L R_5 R_L g_m + 2 C_4 C_L R_L) + s (2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + 2 C_L R_L g_m + C_L)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{2 \sqrt{C_4} \sqrt{C_L} R_5 \sqrt{R_L g_m} \sqrt{\frac{g_m}{R_5 g_m + 1}} + 2 \sqrt{C_4} \sqrt{C_L} \sqrt{R_L} \sqrt{\frac{g_m}{R_5 g_m + 1}}}{2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + 2 C_L R_L g_m + C_L} \\
\text{wo: } & \sqrt{\frac{g_m}{C_4 C_L R_5 R_L g_m + C_4 C_L R_L}} \\
\text{bandwidth: } & \frac{\sqrt{\frac{g_m}{C_4 C_L R_5 R_L g_m + C_4 C_L R_L}} (2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + 2 C_L R_L g_m + C_L)}{2 \sqrt{C_4} \sqrt{C_L} R_5 \sqrt{R_L g_m} \sqrt{\frac{g_m}{R_5 g_m + 1}} + 2 \sqrt{C_4} \sqrt{C_L} \sqrt{R_L} \sqrt{\frac{g_m}{R_5 g_m + 1}}} \\
\text{K-LP: } & \frac{R_5 g_m - 1}{2 g_m} \\
\text{K-HP: } & 0 \\
\text{K-BP: } & \frac{C_L R_5 R_L g_m - C_L R_L}{2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + 2 C_L R_L g_m + C_L} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{-C_5 R_L s + R_L g_m}{2 C_4 C_5 R_L s^2 + g_m + s (2 C_4 R_L g_m + 2 C_5 R_L g_m + C_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{2} \sqrt{C_4} \sqrt{C_5} \sqrt{R_L} \sqrt{g_m}}{2 C_4 R_L g_m + 2 C_5 R_L g_m + C_5} \\
\text{wo: } & \frac{\sqrt{2} \sqrt{g_m}}{2 \sqrt{C_4} \sqrt{C_5} \sqrt{R_L}} \\
\text{bandwidth: } & \frac{2 C_4 R_L g_m + 2 C_5 R_L g_m + C_5}{2 C_4 C_5 R_L} \\
\text{K-LP: } & R_L \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5 R_L}{2 C_4 R_L g_m + 2 C_5 R_L g_m + C_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_L s + R_L g_m}{g_m + s^2 (2 C_4 C_5 R_L + C_5 C_L R_L) + s (2 C_4 R_L g_m + 2 C_5 R_L g_m + C_5 + C_L R_L g_m)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{\sqrt{C_5} \sqrt{R_L} \sqrt{g_m} \sqrt{2 C_4 + C_L}}{2 C_4 R_L g_m + 2 C_5 R_L g_m + C_5 + C_L R_L g_m} \\
\text{wo: } & \frac{\sqrt{g_m}}{\sqrt{2 C_4 C_5 R_L + C_5 C_L R_L}} \\
\text{bandwidth: } & \frac{2 C_4 R_L g_m + 2 C_5 R_L g_m + C_5 + C_L R_L g_m}{\sqrt{C_5} \sqrt{R_L} \sqrt{2 C_4 + C_L} \sqrt{2 C_4 C_5 R_L + C_5 C_L R_L}} \\
\text{K-LP: } & R_L \\
\text{K-HP: } & 0 \\
\text{K-BP: } & -\frac{C_5 R_L}{2 C_4 R_L g_m + 2 C_5 R_L g_m + C_5 + C_L R_L g_m} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \text{None}
\end{aligned}$$

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$

$$H(s) = \frac{-C_5 R_5 R_L s + R_5 R_L g_m - R_L}{2C_4 C_5 R_5 R_L s^2 + R_5 g_m + 2R_L g_m + s(2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_L}\sqrt{R_5 g_m + 2R_L g_m + 1}}{2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5}$
wo: $\frac{\sqrt{2}\sqrt{R_5 g_m + 2R_L g_m + 1}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_L}}$
bandwidth: $\frac{2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5}{2C_4 C_5 R_5 R_L}$
K-LP: $\frac{R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_5 R_L}{2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5}$
Qz: None
Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m - 1}{2g_m + s^2(2C_4 C_5 R_5 + C_5 C_L R_5) + s(2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_5}\sqrt{R_5}\sqrt{g_m}\sqrt{2C_4 + C_L}}{2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{2C_4 C_5 R_5 + C_5 C_L R_5}}$
bandwidth: $\frac{2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L}{\sqrt{C_5}\sqrt{R_5}\sqrt{2C_4 + C_L}\sqrt{2C_4 C_5 R_5 + C_5 C_L R_5}}$
K-LP: $\frac{R_5 g_m - 1}{2g_m}$
K-HP: 0
K-BP: $-\frac{C_5 R_5}{2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L}$
Qz: None
Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_5 R_L s + R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + s^2(2C_4 C_5 R_5 R_L + C_5 C_L R_5 R_L) + s(2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_5}\sqrt{R_5}\sqrt{R_L}\sqrt{2C_4 + C_L}\sqrt{R_5 g_m + 2R_L g_m + 1}}{2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L}$
wo: $\frac{\sqrt{R_5 g_m + 2R_L g_m + 1}}{\sqrt{2C_4 C_5 R_5 R_L + C_5 C_L R_5 R_L}}$
bandwidth: $\frac{2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L}{\sqrt{C_5}\sqrt{R_5}\sqrt{R_L}\sqrt{2C_4 + C_L}\sqrt{2C_4 C_5 R_5 R_L + C_5 C_L R_5 R_L}}$
K-LP: $\frac{R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + 1}$
K-HP: 0
K-BP: $-\frac{C_5 R_5 R_L}{2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L}$
Qz: None
Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{R_L g_m + s(C_5 R_5 R_L g_m - C_5 R_L)}{g_m + s^2(2C_4 C_5 R_5 R_L g_m + 2C_4 C_5 R_L) + s(2C_4 R_L g_m + C_5 R_5 g_m + 2C_5 R_L g_m + C_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}R_5\sqrt{R_L}g_m\sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_L}\sqrt{\frac{g_m}{R_5 g_m + 1}}}{2C_4 R_L g_m + C_5 R_5 g_m + 2C_5 R_L g_m + C_5}$
wo: $\sqrt{\frac{g_m}{2C_4 C_5 R_5 R_L g_m + 2C_4 C_5 R_L}}$

$$\text{bandwidth: } \frac{\sqrt{\frac{gm}{2C_4C_5R_5R_Lg_m+2C_4C_5R_L}}(2C_4R_Lg_m+C_5R_5g_m+2C_5R_Lg_m+C_5)}{\sqrt{2}\sqrt{C_4}\sqrt{C_5}R_5\sqrt{R_Lg_m}\sqrt{\frac{gm}{R_5g_m+1}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_L}\sqrt{\frac{gm}{R_5g_m+1}}}$$

K-LP: R_L

K-HP: 0

K-BP: $\frac{C_5R_5R_Lg_m-C_5R_L}{2C_4R_Lg_m+C_5R_5g_m+2C_5R_Lg_m+C_5}$

Qz: None

Wz: None

$$\mathbf{8.14 \quad X-INVALID-NUMER-14} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4s}, R_5 + \frac{1}{C_5s}, \frac{R_L}{C_LR_Ls+1} \right)$$

$$H(s) = \frac{R_Lg_m + s(C_5R_5R_Lg_m - C_5R_L)}{g_m + s^2(2C_4C_5R_5R_Lg_m + 2C_4C_5R_L + C_5C_LR_5R_Lg_m + C_5C_LR_L) + s(2C_4R_Lg_m + C_5R_5g_m + 2C_5R_Lg_m + C_5 + C_LR_Lg_m)}$$

Parameters:

$$\text{Q: } \frac{2C_4\sqrt{C_5}R_5\sqrt{R_Lg_m}\sqrt{\frac{gm}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+2C_4\sqrt{C_5}\sqrt{R_L}\sqrt{\frac{gm}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+\sqrt{C_5}C_LR_5\sqrt{R_Lg_m}\sqrt{\frac{gm}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+\sqrt{C_5}C_L\sqrt{R_L}\sqrt{\frac{gm}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}}{2C_4R_Lg_m+C_5R_5g_m+2C_5R_Lg_m+C_5+C_LR_Lg_m}$$

$$\text{wo: } \sqrt{\frac{gm}{2C_4C_5R_5R_Lg_m+2C_4C_5R_L+C_5C_LR_5R_Lg_m+C_5C_LR_L}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{gm}{2C_4C_5R_5R_Lg_m+2C_4C_5R_L+C_5C_LR_5R_Lg_m+C_5C_LR_L}}(2C_4R_Lg_m+C_5R_5g_m+2C_5R_Lg_m+C_5+C_LR_Lg_m)}{2C_4\sqrt{C_5}R_5\sqrt{R_Lg_m}\sqrt{\frac{gm}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+2C_4\sqrt{C_5}\sqrt{R_L}\sqrt{\frac{gm}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+\sqrt{C_5}C_LR_5\sqrt{R_Lg_m}\sqrt{\frac{gm}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+\sqrt{C_5}C_L\sqrt{R_L}\sqrt{\frac{gm}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}}$$

K-LP: R_L

K-HP: 0

K-BP: $\frac{C_5R_5R_Lg_m-C_5R_L}{2C_4R_Lg_m+C_5R_5g_m+2C_5R_Lg_m+C_5+C_LR_Lg_m}$

Qz: None

Wz: None

$$\mathbf{8.15 \quad X-INVALID-NUMER-15} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4R_4s+1}, R_5, R_L + \frac{1}{C_Ls} \right)$$

$$H(s) = \frac{R_4R_5g_m - R_4 + s(C_LR_4R_5R_Lg_m - C_LR_4R_L)}{2R_4g_m + 2R_5g_m + s^2(2C_4C_LR_4R_5R_Lg_m + 2C_4C_LR_4R_L) + s(2C_4R_4R_5g_m + 2C_4R_4 + C_LR_4R_5g_m + 2C_LR_4R_Lg_m + C_LR_4 + 2C_LR_5R_Lg_m + 2C_LR_L) + 2}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}R_5\sqrt{R_Lg_m}\sqrt{\frac{R_4g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}+2\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{R_4g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}}{2C_4R_4R_5g_m+2C_4R_4+C_LR_4R_5g_m+2C_LR_4R_Lg_m+C_LR_4+2C_LR_5R_Lg_m+2C_LR_L}$$

$$\text{wo: } \sqrt{\frac{R_4g_m+R_5g_m+1}{C_4C_LR_4R_5R_Lg_m+C_4C_LR_4R_L}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{R_4g_m+R_5g_m+1}{C_4C_LR_4R_5R_Lg_m+C_4C_LR_4R_L}}(2C_4R_4R_5g_m+2C_4R_4+C_LR_4R_5g_m+2C_LR_4R_Lg_m+C_LR_4+2C_LR_5R_Lg_m+2C_LR_L)}{2\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}R_5\sqrt{R_Lg_m}\sqrt{\frac{R_4g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}+2\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{R_4g_m}{R_5g_m+1}+\frac{R_5g_m}{R_5g_m+1}+\frac{1}{R_5g_m+1}}}$$

K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$

K-HP: 0

K-BP: $\frac{C_LR_4R_5R_Lg_m-C_LR_4R_L}{2C_4R_4R_5g_m+2C_4R_4+C_LR_4R_5g_m+2C_LR_4R_Lg_m+C_LR_4+2C_LR_5R_Lg_m+2C_LR_L}$

Qz: None

Wz: None

$$\mathbf{8.16 \quad X-INVALID-NUMER-16} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4R_4s+1}, \frac{1}{C_5s}, R_L \right)$$

$$H(s) = \frac{-C_5R_4R_Ls + R_4R_Lg_m}{2C_4C_5R_4R_Ls^2 + R_4g_m + 2R_Lg_m + s(2C_4R_4R_Lg_m + 2C_5R_4R_Lg_m + C_5R_4 + 2C_5R_L)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_L}\sqrt{g_m}\sqrt{R_4+2R_L}}{2C_4R_4R_Lg_m+2C_5R_4R_Lg_m+C_5R_4+2C_5R_L}$$

$$\text{wo: } \frac{\sqrt{2}\sqrt{R_4g_m+2R_Lg_m}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_L}}$$

$$\text{bandwidth: } \frac{\sqrt{R_4g_m+2R_Lg_m}(2C_4R_4R_Lg_m+2C_5R_4R_Lg_m+C_5R_4+2C_5R_L)}{2C_4C_5R_4R_L\sqrt{g_m}\sqrt{R_4+2R_L}}$$

K-LP: $\frac{R_4R_L}{R_4+2R_L}$

K-HP: 0

K-BP: $-\frac{C_5R_4R_L}{2C_4R_4R_Lg_m+2C_5R_4R_Lg_m+C_5R_4+2C_5R_L}$

Qz: None

Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_4 s + R_4 g_m}{2g_m + s^2 (2C_4 C_5 R_4 + C_5 C_L R_4) + s (2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 + C_L R_4 g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_5}\sqrt{R_4}\sqrt{g_m}\sqrt{2C_4+C_L}}{2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 + C_L R_4 g_m} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{g_m}}{\sqrt{2C_4 C_5 R_4 + C_5 C_L R_4}} \\ \text{bandwidth: } & \frac{2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 + C_L R_4 g_m}{\sqrt{C_5}\sqrt{R_4}\sqrt{2C_4+C_L}\sqrt{2C_4 C_5 R_4 + C_5 C_L R_4}} \\ \text{K-LP: } & \frac{R_4}{2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_4}{2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 + C_L R_4 g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.18 X-INVALID-NUMER-18 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_4 R_L s + R_4 R_L g_m}{R_4 g_m + 2R_L g_m + s^2 (2C_4 C_5 R_4 R_L + C_5 C_L R_4 R_L) + s (2C_4 R_4 R_L g_m + 2C_5 R_4 R_L g_m + C_5 R_4 + 2C_5 R_L + C_L R_4 R_L g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_5}\sqrt{R_4}\sqrt{R_L}\sqrt{g_m}\sqrt{2C_4+C_L}\sqrt{R_4+2R_L}}{2C_4 R_4 R_L g_m + 2C_5 R_4 R_L g_m + C_5 R_4 + 2C_5 R_L + C_L R_4 R_L g_m} \\ \text{wo: } & \frac{\sqrt{R_4 g_m + 2R_L g_m}}{\sqrt{2C_4 C_5 R_4 R_L + C_5 C_L R_4 R_L}} \\ \text{bandwidth: } & \frac{\sqrt{R_4 g_m + 2R_L g_m} (2C_4 R_4 R_L g_m + 2C_5 R_4 R_L g_m + C_5 R_4 + 2C_5 R_L + C_L R_4 R_L g_m)}{\sqrt{C_5}\sqrt{R_4}\sqrt{R_L}\sqrt{g_m}\sqrt{2C_4+C_L}\sqrt{R_4+2R_L}\sqrt{2C_4 C_5 R_4 R_L + C_5 C_L R_4 R_L}} \\ \text{K-LP: } & \frac{R_4 R_L}{R_4 + 2R_L} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_4 R_L}{2C_4 R_4 R_L g_m + 2C_5 R_4 R_L g_m + C_5 R_4 + 2C_5 R_L + C_L R_4 R_L g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.19 X-INVALID-NUMER-19 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$

$$H(s) = \frac{-C_5 R_4 R_5 R_L s + R_4 R_5 R_L g_m - R_4 R_L}{2C_4 C_5 R_4 R_5 R_L s^2 + R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L + s (2C_4 R_4 R_5 R_L g_m + 2C_4 R_4 R_L + 2C_5 R_4 R_5 R_L g_m + C_5 R_4 R_5 + 2C_5 R_5 R_L)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{R_L}\sqrt{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L}}{2C_4 R_4 R_5 R_L g_m + 2C_4 R_4 R_L + 2C_5 R_4 R_5 R_L g_m + C_5 R_4 R_5 + 2C_5 R_5 R_L} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{R_L}} \\ \text{bandwidth: } & \frac{2C_4 R_4 R_5 R_L g_m + 2C_4 R_4 R_L + 2C_5 R_4 R_5 R_L g_m + C_5 R_4 R_5 + 2C_5 R_5 R_L}{2C_4 C_5 R_4 R_5 R_L} \\ \text{K-LP: } & \frac{R_4 R_5 R_L g_m - R_4 R_L}{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_4 R_5 R_L}{2C_4 R_4 R_5 R_L g_m + 2C_4 R_4 R_L + 2C_5 R_4 R_5 R_L g_m + C_5 R_4 R_5 + 2C_5 R_5 R_L} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.20 X-INVALID-NUMER-20 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s^2 (2C_4 C_5 R_4 R_5 + C_5 C_L R_4 R_5) + s (2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5 + C_L R_4 R_5 g_m + C_L R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{2C_4+C_L}\sqrt{R_4 g_m + R_5 g_m + 1}}{2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5 + C_L R_4 R_5 g_m + C_L R_4} \\ \text{wo: } & \frac{\sqrt{2R_4 g_m + 2R_5 g_m + 2}}{\sqrt{2C_4 C_5 R_4 R_5 + C_5 C_L R_4 R_5}} \end{aligned}$$

bandwidth: $\frac{\sqrt{2}\sqrt{2R_4g_m+2R_5g_m+2}(2C_4R_4R_5g_m+2C_4R_4+2C_5R_4R_5g_m+2C_5R_5+C_LR_4R_5g_m+C_LR_4)}{2\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{2C_4+C_L}\sqrt{2C_4C_5R_4R_5+C_5C_LR_4R_5}\sqrt{R_4g_m+R_5g_m+1}}$
K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$
K-HP: 0
K-BP: $-\frac{C_5R_4R_5}{2C_4R_4R_5g_m+2C_4R_4+2C_5R_4R_5g_m+2C_5R_5+C_LR_4R_5g_m+C_LR_4}$
Qz: None
Wz: None

8.21 X-INVALID-NUMER-21 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4R_4s+1}, \frac{R_5}{C_5R_5s+1}, \frac{R_L}{C_LR_Ls+1} \right)$

$$H(s) = \frac{-C_5R_4R_5R_Ls + R_4R_5R_Lg_m - R_4R_L}{R_4R_5g_m + 2R_4R_Lg_m + R_4 + 2R_5R_Lg_m + 2R_L + s^2(2C_4C_5R_4R_5R_L + C_5C_LR_4R_5R_L) + s(2C_4R_4R_5R_Lg_m + 2C_4R_4R_L + 2C_5R_4R_5R_Lg_m + C_5R_4R_5 + 2C_5R_5R_L + C_LR_4R_5R_Lg_m + C_LR_4R_L)}$$

Parameters:

Q: $\frac{\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{R_L}\sqrt{2C_4+C_L}\sqrt{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}{2C_4R_4R_5R_Lg_m+2C_4R_4R_L+2C_5R_4R_5R_Lg_m+C_5R_4R_5+2C_5R_5R_L+C_LR_4R_5R_Lg_m+C_LR_4R_L}$
wo: $\frac{\sqrt{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}{\sqrt{2C_4C_5R_4R_5R_L+C_5C_LR_4R_5R_L}}$
bandwidth: $\frac{2C_4R_4R_5R_Lg_m+2C_4R_4R_L+2C_5R_4R_5R_Lg_m+C_5R_4R_5+2C_5R_5R_L+C_LR_4R_5R_Lg_m+C_LR_4R_L}{\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{R_L}\sqrt{2C_4+C_L}\sqrt{2C_4C_5R_4R_5R_L+C_5C_LR_4R_5R_L}}$
K-LP: $\frac{R_4R_5R_Lg_m-R_4R_L}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}$
K-HP: 0
K-BP: $-\frac{C_5R_4R_5R_L}{2C_4R_4R_5R_Lg_m+2C_4R_4R_L+2C_5R_4R_5R_Lg_m+C_5R_4R_5+2C_5R_5R_L+C_LR_4R_5R_Lg_m+C_LR_4R_L}$
Qz: None
Wz: None

8.22 X-INVALID-NUMER-22 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4R_4s+1}, R_5 + \frac{1}{C_5s}, R_L \right)$

$$H(s) = \frac{R_4R_Lg_m + s(C_5R_4R_5R_Lg_m - C_5R_4R_L)}{R_4g_m + 2R_Lg_m + s^2(2C_4C_5R_4R_5R_Lg_m + 2C_4C_5R_4R_L) + s(2C_4R_4R_Lg_m + C_5R_4R_5g_m + 2C_5R_4R_Lg_m + C_5R_4 + 2C_5R_5R_Lg_m + 2C_5R_L)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}R_5\sqrt{R_L}g_m\sqrt{\frac{g_m}{R_5g_m+1}}\sqrt{R_4+2R_L}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{g_m}{R_5g_m+1}}\sqrt{R_4+2R_L}}{2C_4R_4R_Lg_m+C_5R_4R_5g_m+2C_5R_4R_Lg_m+C_5R_4+2C_5R_5R_Lg_m+2C_5R_L}$
wo: $\sqrt{\frac{R_4g_m+2R_Lg_m}{2C_4C_5R_4R_5R_Lg_m+2C_4C_5R_4R_L}}$
bandwidth: $\frac{\sqrt{\frac{R_4g_m+2R_Lg_m}{2C_4C_5R_4R_5R_Lg_m+2C_4C_5R_4R_L}}(2C_4R_4R_Lg_m+C_5R_4R_5g_m+2C_5R_4R_Lg_m+C_5R_4+2C_5R_5R_Lg_m+2C_5R_L)}{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}R_5\sqrt{R_L}g_m\sqrt{\frac{g_m}{R_5g_m+1}}\sqrt{R_4+2R_L}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_L}\sqrt{\frac{g_m}{R_5g_m+1}}\sqrt{R_4+2R_L}}$
K-LP: $\frac{R_4R_L}{R_4+2R_L}$
K-HP: 0
K-BP: $\frac{C_5R_4R_5R_Lg_m-C_5R_4R_L}{2C_4R_4R_Lg_m+C_5R_4R_5g_m+2C_5R_4R_Lg_m+C_5R_4+2C_5R_5R_Lg_m+2C_5R_L}$
Qz: None
Wz: None

8.23 X-INVALID-NUMER-23 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4R_4s+1}, R_5 + \frac{1}{C_5s}, \frac{1}{C_Ls} \right)$

$$H(s) = \frac{R_4g_m + s(C_5R_4R_5g_m - C_5R_4)}{2g_m + s^2(2C_4C_5R_4R_5g_m + 2C_4C_5R_4 + C_5C_LR_4R_5g_m + C_5C_LR_4) + s(2C_4R_4g_m + 2C_5R_4g_m + 2C_5R_5g_m + 2C_5 + C_LR_4g_m)}$$

Parameters:

Q: $\frac{2\sqrt{2}C_4\sqrt{C_5}\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+2\sqrt{2}C_4\sqrt{C_5}\sqrt{R_4}\sqrt{\frac{g_m}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+\sqrt{2}\sqrt{C_5}C_L\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+\sqrt{2}\sqrt{C_5}C_L\sqrt{R_4}\sqrt{\frac{g_m}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}}{2C_4R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5+C_LR_4g_m}$
wo: $\sqrt{2}\sqrt{\frac{g_m}{2C_4C_5R_4R_5g_m+2C_4C_5R_4+C_5C_LR_4R_5g_m+C_5C_LR_4}}$
bandwidth: $\frac{2\sqrt{2}\sqrt{\frac{g_m}{2C_4C_5R_4R_5g_m+2C_4C_5R_4+C_5C_LR_4R_5g_m+C_5C_LR_4}}(2C_4R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5+C_LR_4g_m)}{2\sqrt{2}C_4\sqrt{C_5}\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+2\sqrt{2}C_4\sqrt{C_5}\sqrt{R_4}\sqrt{\frac{g_m}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+\sqrt{2}\sqrt{C_5}C_L\sqrt{R_4}R_5g_m\sqrt{\frac{g_m}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}+\sqrt{2}\sqrt{C_5}C_L\sqrt{R_4}\sqrt{\frac{g_m}{2C_4R_5g_m+2C_4+C_LR_5g_m+C_L}}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $\frac{C_5R_4R_5g_m-C_5R_4}{2C_4R_4g_m+2C_5R_4g_m+2C_5R_5g_m+2C_5+C_LR_4g_m}$
Qz: None
Wz: None

8.24 X-INVALID-NUMER-24 $Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_4 R_L g_m + s (C_5 R_4 R_5 R_L g_m - C_5 R_4 R_L)}{R_4 g_m + 2 R_L g_m + s^2 (2 C_4 C_5 R_4 R_5 R_L g_m + 2 C_4 C_5 R_4 R_L + C_5 C_L R_4 R_5 R_L g_m + C_5 C_L R_4 R_L) + s (2 C_4 R_4 R_L g_m + C_5 R_4 R_5 g_m + 2 C_5 R_4 R_L g_m + C_5 R_4 + 2 C_5 R_5 R_L g_m + 2 C_5 R_L + C_L R_4 R_L g_m)}$$

Parameters:

Q: $\frac{2 C_4 \sqrt{C_5} \sqrt{R_4} R_5 \sqrt{R_L} g_m \sqrt{\frac{g_m}{2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L}} \sqrt{R_4 + 2 R_L} + 2 C_4 \sqrt{C_5} \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{g_m}{2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L}} \sqrt{R_4 + 2 R_L} + \sqrt{C_5} C_L \sqrt{R_4} R_5 \sqrt{R_L} g_m \sqrt{\frac{g_m}{2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L}} \sqrt{R_4 + 2 R_L} + \sqrt{C_5} C_L \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{g_m}{2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L}} \sqrt{R_4 + 2 R_L}}{2 C_4 R_4 R_L g_m + C_5 R_4 R_5 g_m + 2 C_5 R_4 R_L g_m + C_5 R_4 + 2 C_5 R_5 R_L g_m + 2 C_5 R_L + C_L R_4 R_L g_m}$

wo: $\sqrt{\frac{R_4 g_m + 2 R_L g_m}{2 C_4 C_5 R_4 R_5 R_L g_m + 2 C_4 C_5 R_4 R_L + C_5 C_L R_4 R_5 R_L g_m + C_5 C_L R_4 R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_4 g_m + 2 R_L g_m}{2 C_4 C_5 R_4 R_5 R_L g_m + 2 C_4 C_5 R_4 R_L + C_5 C_L R_4 R_5 R_L g_m + C_5 C_L R_4 R_L}} (2 C_4 R_4 R_L g_m + C_5 R_4 R_5 g_m + 2 C_5 R_4 R_L g_m + C_5 R_4 + 2 C_5 R_5 R_L g_m + 2 C_5 R_L + C_L R_4 R_L g_m)}{2 C_4 \sqrt{C_5} \sqrt{R_4} R_5 \sqrt{R_L} g_m \sqrt{\frac{g_m}{2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L}} \sqrt{R_4 + 2 R_L} + 2 C_4 \sqrt{C_5} \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{g_m}{2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L}} \sqrt{R_4 + 2 R_L} + \sqrt{C_5} C_L \sqrt{R_4} R_5 \sqrt{R_L} g_m \sqrt{\frac{g_m}{2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L}} \sqrt{R_4 + 2 R_L} + \sqrt{C_5} C_L \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{g_m}{2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L}} \sqrt{R_4 + 2 R_L}}$

K-LP: $\frac{R_4 R_L}{R_4 + 2 R_L}$

K-HP: 0

K-BP: $\frac{C_5 R_4 R_5 R_L g_m - C_5 R_4 R_L}{2 C_4 R_4 R_L g_m + C_5 R_4 R_5 g_m + 2 C_5 R_4 R_L g_m + C_5 R_4 + 2 C_5 R_5 R_L g_m + 2 C_5 R_L + C_L R_4 R_L g_m}$

Qz: None

Wz: None

8.25 X-INVALID-NUMER-25 $Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_5 g_m + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2 g_m + s^2 (C_4 C_L R_4 R_5 g_m + C_4 C_L R_4) + s (2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} R_5 g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{2} \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} \sqrt{\frac{g_m}{R_5 g_m + 1}}}{2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L}$

wo: $\sqrt{2} \sqrt{\frac{g_m}{C_4 C_L R_4 R_5 g_m + C_4 C_L R_4}}$

bandwidth: $\frac{\sqrt{2} \sqrt{\frac{g_m}{C_4 C_L R_4 R_5 g_m + C_4 C_L R_4}} (2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L)}{\sqrt{2} \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} R_5 g_m \sqrt{\frac{g_m}{R_5 g_m + 1}} + \sqrt{2} \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} \sqrt{\frac{g_m}{R_5 g_m + 1}}}$

K-LP: $\frac{R_5 g_m - 1}{2 g_m}$

K-HP: 0

K-BP: $\frac{C_4 R_4 R_5 g_m - C_4 R_4}{2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4 + C_L R_5 g_m + C_L}$

Qz: None

Wz: None

8.26 X-INVALID-NUMER-26 $Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_5 R_L g_m - R_L + s (C_4 R_4 R_5 R_L g_m - C_4 R_4 R_L)}{R_5 g_m + 2 R_L g_m + s^2 (C_4 C_L R_4 R_5 R_L g_m + C_4 C_L R_4 R_L) + s (C_4 R_4 R_5 g_m + 2 C_4 R_4 R_L g_m + C_4 R_4 + 2 C_4 R_5 R_L g_m + 2 C_4 R_L + C_L R_5 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_4} \sqrt{C_L} \sqrt{R_4} R_5 \sqrt{R_L} g_m \sqrt{\frac{R_5 g_m}{R_5 g_m + 1} + \frac{2 R_L g_m}{R_5 g_m + 1} + \frac{1}{R_5 g_m + 1}} + \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{R_5 g_m}{R_5 g_m + 1} + \frac{2 R_L g_m}{R_5 g_m + 1} + \frac{1}{R_5 g_m + 1}}}{C_4 R_4 R_5 g_m + 2 C_4 R_4 R_L g_m + C_4 R_4 + 2 C_4 R_5 R_L g_m + 2 C_4 R_L + C_L R_5 R_L g_m + C_L R_L}$

wo: $\sqrt{\frac{R_5 g_m + 2 R_L g_m + 1}{C_4 C_L R_4 R_5 R_L g_m + C_4 C_L R_4 R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_5 g_m + 2 R_L g_m + 1}{C_4 C_L R_4 R_5 R_L g_m + C_4 C_L R_4 R_L}} (C_4 R_4 R_5 g_m + 2 C_4 R_4 R_L g_m + C_4 R_4 + 2 C_4 R_5 R_L g_m + 2 C_4 R_L + C_L R_5 R_L g_m + C_L R_L)}{\sqrt{C_4} \sqrt{C_L} \sqrt{R_4} R_5 \sqrt{R_L} g_m \sqrt{\frac{R_5 g_m}{R_5 g_m + 1} + \frac{2 R_L g_m}{R_5 g_m + 1} + \frac{1}{R_5 g_m + 1}} + \sqrt{C_4} \sqrt{C_L} \sqrt{R_4} \sqrt{R_L} \sqrt{\frac{R_5 g_m}{R_5 g_m + 1} + \frac{2 R_L g_m}{R_5 g_m + 1} + \frac{1}{R_5 g_m + 1}}}$

K-LP: $\frac{R_5 R_L g_m - R_L}{R_5 g_m + 2 R_L g_m + 1}$

K-HP: 0

K-BP: $\frac{C_4 R_4 R_5 R_L g_m - C_4 R_4 R_L}{C_4 R_4 R_5 g_m + 2 C_4 R_4 R_L g_m + C_4 R_4 + 2 C_4 R_5 R_L g_m + 2 C_4 R_L + C_L R_5 R_L g_m + C_L R_L}$

Qz: None

Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (\infty, \infty, \infty, R_4, R_5, R_L)$

$$H(s) = \frac{R_4 R_5 R_L g_m - R_4 R_L}{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty, \infty, \infty, R_4, R_5, \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s(C_L R_4 R_5 g_m + C_L R_4) + 2}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty, \infty, \infty, R_4, R_5, \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_4 R_5 R_L g_m - R_4 R_L}{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L + s(C_L R_4 R_5 R_L g_m + C_L R_4 R_L)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty, \infty, \infty, R_4, R_5, R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s(C_L R_4 R_5 R_L g_m - C_L R_4 R_L)}{2R_4 g_m + 2R_5 g_m + s(C_L R_4 R_5 g_m + 2C_L R_4 R_L g_m + C_L R_4 + 2C_L R_5 R_L g_m + 2C_L R_L) + 2}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{1}{C_5 s}, R_L\right)$

$$H(s) = \frac{-C_5 R_4 R_L s + R_4 R_L g_m}{R_4 g_m + 2R_L g_m + s(2C_5 R_4 R_L g_m + C_5 R_4 + 2C_5 R_L)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, R_L\right)$

$$H(s) = \frac{-C_5 R_4 R_5 R_L s + R_4 R_5 R_L g_m - R_4 R_L}{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L + s(2C_5 R_4 R_5 R_L g_m + C_5 R_4 R_5 + 2C_5 R_5 R_L)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, R_L\right)$

$$H(s) = \frac{R_4 R_L g_m + s(C_5 R_4 R_5 R_L g_m - C_5 R_4 R_L)}{R_4 g_m + 2R_L g_m + s(C_5 R_4 R_5 g_m + 2C_5 R_4 R_L g_m + C_5 R_4 + 2C_5 R_5 R_L g_m + 2C_5 R_L)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5, R_L\right)$

$$H(s) = \frac{R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + s(2C_4 R_5 R_L g_m + 2C_4 R_L) + 1}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5, \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_5 g_m - 1}{2g_m + s(2C_4 R_5 g_m + 2C_4 + C_L R_5 g_m + C_L)}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5, \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + s(2C_4 R_5 R_L g_m + 2C_4 R_L + C_L R_5 R_L g_m + C_L R_L) + 1}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 s + g_m}{s^2 (2C_4 C_5 + C_5 C_L) + s (2C_4 g_m + 2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_L s^2 + g_m + s (-C_5 + C_L R_L g_m)}{2C_4 C_5 C_L R_L s^3 + s^2 (2C_4 C_5 + 2C_4 C_L R_L g_m + 2C_5 C_L R_L g_m + C_5 C_L) + s (2C_4 g_m + 2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_5 R_L s^2 + R_5 g_m + s (-C_5 R_5 + C_L R_5 R_L g_m - C_L R_L) - 1}{2C_4 C_5 C_L R_5 R_L s^3 + 2g_m + s^2 (2C_4 C_5 R_5 + 2C_4 C_L R_5 R_L g_m + 2C_4 C_L R_L + 2C_5 C_L R_5 R_L g_m + C_5 C_L R_5) + s (2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m + C_L R_5 g_m + 2C_L R_L g_m + C_L)}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m + s (C_5 R_5 g_m - C_5)}{s^2 (2C_4 C_5 R_5 g_m + 2C_4 C_5 + C_5 C_L R_5 g_m + C_5 C_L) + s (2C_4 g_m + 2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\infty, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m + s^2 (C_5 C_L R_5 R_L g_m - C_5 C_L R_L) + s (C_5 R_5 g_m - C_5 + C_L R_L g_m)}{s^3 (2C_4 C_5 C_L R_5 R_L g_m + 2C_4 C_5 C_L R_L) + s^2 (2C_4 C_5 R_5 g_m + 2C_4 C_5 + 2C_4 C_L R_L g_m + C_5 C_L R_5 g_m + 2C_5 C_L R_L g_m + C_5 C_L) + s (2C_4 g_m + 2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, R_L \right)$$

$$H(s) = \frac{R_4 R_5 R_L g_m - R_4 R_L}{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L + s (2C_4 R_4 R_5 R_L g_m + 2C_4 R_4 R_L)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s (2C_4 R_4 R_5 g_m + 2C_4 R_4 + C_L R_4 R_5 g_m + C_L R_4) + 2}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_4 R_5 R_L g_m - R_4 R_L}{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L + s (2C_4 R_4 R_5 R_L g_m + 2C_4 R_4 R_L + C_L R_4 R_5 R_L g_m + C_L R_4 R_L)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_4 R_L s^2 + R_4 g_m + s (-C_5 R_4 + C_L R_4 R_L g_m)}{2C_4 C_5 C_L R_4 R_L s^3 + 2g_m + s^2 (2C_4 C_5 R_4 + 2C_4 C_L R_4 R_L g_m + 2C_5 C_L R_4 R_L g_m + C_5 C_L R_4 + 2C_5 C_L R_L) + s (2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 + C_L R_4 g_m + 2C_L R_L g_m)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\infty, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_4 R_5 R_L s^2 + R_4 R_5 g_m - R_4 + s (-C_5 R_4 R_5 + C_L R_4 R_5 R_L g_m - C_L R_4 R_L)}{2C_4 C_5 C_L R_4 R_5 R_L s^3 + 2R_4 g_m + 2R_5 g_m + s^2 (2C_4 C_5 R_4 R_5 + 2C_4 C_L R_4 R_5 R_L g_m + 2C_4 C_L R_4 R_L + 2C_5 C_L R_4 R_5 R_L g_m + C_5 C_L R_4 R_5 + 2C_5 C_L R_5 R_L) + s (2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_5 + C_L R_4 R_5 g_m + 2C_L R_4 R_L g_m + C_L R_4 + 2C_L R_5 R_L g_m + 2C_L R_5)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\infty, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_4 g_m + s^2 (C_5 C_L R_4 R_5 R_L g_m - C_5 C_L R_4 R_L) + s (C_5 R_4 R_5 g_m - C_5 R_4 + C_L R_4 R_L g_m)}{2g_m + s^3 (2C_4 C_5 C_L R_4 R_5 R_L g_m + 2C_4 C_5 C_L R_4 R_L) + s^2 (2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4 + 2C_4 C_L R_4 R_L g_m + C_5 C_L R_4 R_5 g_m + 2C_5 C_L R_4 R_L g_m + C_5 C_L R_4 + 2C_5 C_L R_5 R_L g_m + 2C_5 C_L R_L) + s (2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5 + C_L R_4 g_m + 2C_L R_L g_m)}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\infty, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad R_L \right)$$

$$H(s) = \frac{R_5 R_L g_m - R_L + s (C_4 R_4 R_5 R_L g_m - C_4 R_4 R_L)}{R_5 g_m + 2R_L g_m + s (C_4 R_4 R_5 g_m + 2C_4 R_4 R_L g_m + C_4 R_4 + 2C_4 R_5 R_L g_m + 2C_4 R_L) + 1}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\infty, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 s^2 + g_m + s (C_4 R_4 g_m - C_5)}{C_4 C_5 C_L R_4 s^3 + s^2 (2C_4 C_5 R_4 g_m + 2C_4 C_5 + C_4 C_L R_4 g_m + C_5 C_L) + s (2C_4 g_m + 2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\infty, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 R_L s^2 + R_L g_m + s (C_4 R_4 R_L g_m - C_5 R_L)}{C_4 C_5 C_L R_4 R_L s^3 + g_m + s^2 (2C_4 C_5 R_4 R_L g_m + C_4 C_5 R_4 + 2C_4 C_5 R_L + C_4 C_L R_4 R_L g_m + C_5 C_L R_L) + s (C_4 R_4 g_m + 2C_4 R_L g_m + 2C_5 R_L g_m + C_5 + C_L R_L g_m)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\infty, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_4 C_5 C_L R_4 R_L s^3 + g_m + s^2 (-C_4 C_5 R_4 + C_4 C_L R_4 R_L g_m - C_5 C_L R_L) + s (C_4 R_4 g_m - C_5 + C_L R_L g_m)}{s^3 (2C_4 C_5 C_L R_4 R_L g_m + C_4 C_5 C_L R_4 + 2C_4 C_5 C_L R_L) + s^2 (2C_4 C_5 R_4 g_m + 2C_4 C_5 + C_4 C_L R_4 g_m + 2C_4 C_L R_L g_m + 2C_5 C_L R_L g_m + C_5 C_L) + s (2C_4 g_m + 2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\infty, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 R_5 s^2 + R_5 g_m + s (C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{C_4 C_5 C_L R_4 R_5 s^3 + 2g_m + s^2 (2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5 + C_4 C_L R_4 R_5 g_m + C_4 C_L R_4 + C_5 C_L R_5) + s (2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\infty, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 R_5 R_L s^2 + R_5 R_L g_m - R_L + s (C_4 R_4 R_5 R_L g_m - C_4 R_4 R_L - C_5 R_5 R_L)}{C_4 C_5 C_L R_4 R_5 R_L s^3 + R_5 g_m + 2R_L g_m + s^2 (2C_4 C_5 R_4 R_5 R_L g_m + C_4 C_5 R_4 R_5 + 2C_4 C_5 R_5 R_L + C_4 C_L R_4 R_5 R_L g_m + C_4 C_L R_4 R_L + C_5 C_L R_5 R_L) + s (C_4 R_4 R_5 g_m + 2C_4 R_4 R_L g_m + C_4 R_4 + 2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5 + C_L R_5 R_L g_m + C_L R_L) + 1}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\infty, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_4 C_5 C_L R_4 R_5 R_L s^3 + R_5 g_m + s^2 (-C_4 C_5 R_4 R_5 + C_4 C_L R_4 R_5 R_L g_m - C_4 C_L R_4 R_L - C_5 C_L R_5 R_L) + s (C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5 + C_L R_5 R_L g_m - C_L R_L) - 1}{2g_m + s^3 (2C_4 C_5 C_L R_4 R_5 R_L g_m + C_4 C_5 C_L R_4 R_5 + 2C_4 C_5 C_L R_5 R_L) + s^2 (2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5 + C_4 C_L R_4 R_5 g_m + 2C_4 C_L R_4 R_L g_m + C_4 C_L R_4 + 2C_4 C_L R_5 R_L g_m + 2C_4 C_L R_L + 2C_5 C_L R_5 R_L g_m + C_5 C_L R_5) + s (2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m + C_L R_5 g_m + C_L)}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(\infty, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^3 (C_4 C_5 C_L R_4 R_5 g_m + C_4 C_5 C_L R_4) + s^2 (2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5 + C_4 C_L R_4 g_m + C_5 C_L R_5 g_m + C_5 C_L) + s (2C_4 g_m + 2C_5 g_m + C_L g_m)}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(\infty, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L g_m + s^2 (C_4 C_5 R_4 R_5 R_L g_m - C_4 C_5 R_4 R_L) + s (C_4 R_4 R_L g_m + C_5 R_5 R_L g_m - C_5 R_L)}{g_m + s^3 (C_4 C_5 C_L R_4 R_5 R_L g_m + C_4 C_5 C_L R_4 R_L) + s^2 (C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4 R_L g_m + C_4 C_5 R_4 + 2C_4 C_5 R_5 R_L g_m + 2C_4 C_5 R_L + C_4 C_L R_4 R_L g_m + C_5 C_L R_5 R_L g_m + C_5 C_L R_L) + s (C_4 R_4 g_m + 2C_4 R_L g_m + C_5 R_5 g_m + 2C_5 R_L g_m + C_5 + C_L R_L g_m)}$$

9.31 X-INVALID-ORDER-31 $Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m + s^3 (C_4 C_5 C_L R_4 R_5 R_L g_m - C_4 C_5 C_L R_4 R_L) + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_4 C_L R_4 R_L g_m + C_5 C_L R_5 R_L g_m - C_5 C_L R_L) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5 + C_L R_L g_m)}{s^3 (C_4 C_5 C_L R_4 R_5 g_m + 2 C_4 C_5 C_L R_4 R_L g_m + C_4 C_5 C_L R_4 + 2 C_4 C_5 C_L R_5 R_L g_m + 2 C_4 C_5 C_L R_L) + s^2 (2 C_4 C_5 R_4 g_m + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5 + C_4 C_L R_4 g_m + 2 C_4 C_L R_L g_m + C_5 C_L R_5 g_m + 2 C_5 C_L R_L g_m + C_5 C_L) + s (2 C_4 g_m + 2 C_5 g_m + C_L g_m)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 C_L R_4 R_L s^2 + R_4 g_m + s (-C_5 R_4 + C_L R_4 R_L g_m)}{2 g_m + s^2 (2 C_5 C_L R_4 R_L g_m + C_5 C_L R_4 + 2 C_5 C_L R_L) + s (2 C_5 R_4 g_m + 2 C_5 + C_L R_4 g_m + 2 C_L R_L g_m)}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4R_Lg_m\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}+\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}+2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_L\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}}{2C_5R_4g_m+2C_5+C_LR_4g_m+2C_LR_Lg_m}$

wo: $\sqrt{2}\sqrt{\frac{g_m}{2C_5C_LR_4R_Lg_m+C_5C_LR_4+2C_5C_LR_L}}$

bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{2C_5C_LR_4R_Lg_m+C_5C_LR_4+2C_5C_LR_L}}(2C_5R_4g_m+2C_5+C_LR_4g_m+2C_LR_Lg_m)}{2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4R_Lg_m\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}+\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}+2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_L\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}}$

K-LP: $\frac{R_4}{2}$

K-HP: $-\frac{R_4R_L}{2R_4R_Lg_m+R_4+2R_L}$

K-BP: $\frac{-C_5R_4+C_LR_4R_Lg_m}{2C_5R_4g_m+2C_5+C_LR_4g_m+2C_LR_Lg_m}$

Qz: None

Wz: $\frac{\sqrt{-g_m}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_L}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(\infty, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 C_L R_4 R_5 R_L s^2 + R_4 R_5 g_m - R_4 + s (-C_5 R_4 R_5 + C_L R_4 R_5 R_L g_m - C_L R_4 R_L)}{2 R_4 g_m + 2 R_5 g_m + s^2 (2 C_5 C_L R_4 R_5 R_L g_m + C_5 C_L R_4 R_5 + 2 C_5 C_L R_5 R_L) + s (2 C_5 R_4 R_5 g_m + 2 C_5 R_5 + C_L R_4 R_5 g_m + 2 C_L R_4 R_L g_m + C_L R_4 + 2 C_L R_5 R_L g_m + 2 C_L R_L) + 2}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4\sqrt{R_5R_Lg_m}\sqrt{\frac{R_4g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}+\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}+2\sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}R_L\sqrt{\frac{R_4g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}}{2C_5R_4R_5g_m+2C_5R_5+C_LR_4R_5g_m+2C_LR_4R_Lg_m+C_LR_4+2C_LR_5R_Lg_m+2C_LR_L}$

wo: $\sqrt{\frac{2R_4g_m+2R_5g_m+2}{2C_5C_LR_4R_5R_Lg_m+C_5C_LR_4R_5+2C_5C_LR_5R_L}}$

bandwidth: $\frac{\sqrt{\frac{2R_4g_m+2R_5g_m+2}{2C_5C_LR_4R_5R_Lg_m+C_5C_LR_4R_5+2C_5C_LR_5R_L}}(2C_5R_4R_5g_m+2C_5R_5+C_LR_4R_5g_m+2C_LR_4R_Lg_m+C_LR_4+2C_LR_5R_Lg_m+2C_LR_L)}{2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4\sqrt{R_5R_Lg_m}\sqrt{\frac{R_4g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}+\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}+2\sqrt{2}\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}R_L\sqrt{\frac{R_4g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}}$

K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$

K-HP: $-\frac{R_4R_L}{2R_4R_Lg_m+R_4+2R_L}$

K-BP: $\frac{-C_5R_4R_5+C_LR_4R_5R_Lg_m-C_LR_4R_L}{2C_5R_4R_5g_m+2C_5R_5+C_LR_4R_5g_m+2C_LR_4R_Lg_m+C_LR_4+2C_LR_5R_Lg_m+2C_LR_L}$

Qz: None

Wz: $\frac{\sqrt{-R_5g_m+1}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_5}\sqrt{R_L}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(\infty, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_4 g_m + s^2 (C_5 C_L R_4 R_5 R_L g_m - C_5 C_L R_4 R_L) + s (C_5 R_4 R_5 g_m - C_5 R_4 + C_L R_4 R_L g_m)}{2 g_m + s^2 (C_5 C_L R_4 R_5 g_m + 2 C_5 C_L R_4 R_L g_m + C_5 C_L R_4 + 2 C_5 C_L R_5 R_L g_m + 2 C_5 C_L R_L) + s (2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5 + C_L R_4 g_m + 2 C_L R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4R_5g_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4R_Lg_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_5R_Lg_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_L\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}}{2C_5R_4g_m+2C_5R_5g_m+2C_5+C_LR_4g_m+2C_LR_Lg_m}$

wo: $\sqrt{2}\sqrt{\frac{g_m}{C_5C_LR_4R_5g_m+2C_5C_LR_4R_Lg_m+C_5C_LR_4+2C_5C_LR_5R_Lg_m+2C_5C_LR_L}}$

bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{C_5C_LR_4R_5g_m+2C_5C_LR_4R_Lg_m+C_5C_LR_4+2C_5C_LR_5R_Lg_m+2C_5C_LR_L}}(2C_5R_4g_m+2C_5R_5g_m+2C_5+C_LR_4g_m+2C_LR_Lg_m)}{\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4R_5g_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4R_Lg_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_4\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_5R_Lg_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_5}\sqrt{C_L}R_L\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}}$

K-LP: $\frac{R_4}{2}$

$$\begin{aligned} \text{K-HP: } & \frac{R_4 R_5 R_L g_m - R_4 R_L}{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L} \\ \text{K-BP: } & \frac{C_5 R_4 R_5 g_m - C_5 R_4 + C_L R_4 R_L g_m}{2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5 + C_L R_4 g_m + 2C_L R_L g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \sqrt{\frac{g_m}{C_5 C_L R_5 R_L g_m - C_5 C_L R_L}} \end{aligned}$$

$$10.4 \quad \mathbf{X-INVALID-WZ-4} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_5 g_m + s^2 (C_4 C_L R_4 R_5 R_L g_m - C_4 C_L R_4 R_L) + s (C_4 R_4 R_5 g_m - C_4 R_4 + C_L R_5 R_L g_m - C_L R_L) - 1}{2g_m + s^2 (C_4 C_L R_4 R_5 g_m + 2C_4 C_L R_4 R_L g_m + C_4 C_L R_4 + 2C_4 C_L R_5 R_L g_m + 2C_4 C_L R_L) + s (2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + C_L R_5 g_m + 2C_L R_L g_m + C_L)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_4R_5g_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_4R_Lg_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_4\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_5R_Lg_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_L\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}}{2C_4R_4g_m+2C_4R_5g_m+2C_4+C_LR_5g_m+2C_LR_Lg_m+C_L}$$

$$\text{wo: } \sqrt{2}\sqrt{\frac{g_m}{C_4C_LR_4R_5g_m+2C_4C_LR_4R_Lg_m+C_4C_LR_4+2C_4C_LR_5R_Lg_m+2C_4C_LR_L}}$$

$$\text{bandwidth: } \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_4R_5g_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_4R_Lg_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_4\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_5R_Lg_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_L\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}}{\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_4R_5g_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_4R_Lg_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_4\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_5R_Lg_m\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}+2\sqrt{2}\sqrt{C_4}\sqrt{C_L}R_L\sqrt{\frac{g_m}{R_4R_5g_m+2R_4R_Lg_m+R_4+2R_5R_Lg_m+2R_L}}}$$

$$\begin{aligned} \text{K-LP: } & \frac{R_5 g_m - 1}{2g_m} \\ \text{K-HP: } & \frac{R_4 R_5 R_L g_m - R_4 R_L}{R_4 R_5 g_m + 2R_4 R_L g_m + R_4 + 2R_5 R_L g_m + 2R_L} \\ \text{K-BP: } & \frac{C_4 R_4 R_5 g_m - C_4 R_4 + C_L R_5 R_L g_m - C_L R_L}{2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + C_L R_5 g_m + 2C_L R_L g_m + C_L} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{1}{\sqrt{C_4}\sqrt{C_L}\sqrt{R_4}\sqrt{R_L}} \end{aligned}$$

$$10.5 \quad \mathbf{X-INVALID-WZ-5} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, R_L \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 R_L s^2 + R_L g_m + s (C_4 R_4 R_L g_m - C_5 R_L)}{g_m + s^2 (2C_4 C_5 R_4 R_L g_m + C_4 C_5 R_4 + 2C_4 C_5 R_L) + s (C_4 R_4 g_m + 2C_4 R_L g_m + 2C_5 R_L g_m + C_5)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_4}\sqrt{C_5}R_4R_Lg_m\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}+2\sqrt{C_4}\sqrt{C_5}R_L\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}}{C_4R_4g_m+2C_4R_Lg_m+2C_5R_Lg_m+C_5}$$

$$\text{wo: } \sqrt{\frac{g_m}{2C_4C_5R_4R_Lg_m+C_4C_5R_4+2C_4C_5R_L}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{g_m}{2C_4C_5R_4R_Lg_m+C_4C_5R_4+2C_4C_5R_L}}(C_4R_4g_m+2C_4R_Lg_m+2C_5R_Lg_m+C_5)}{2\sqrt{C_4}\sqrt{C_5}R_4R_Lg_m\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}+2\sqrt{C_4}\sqrt{C_5}R_L\sqrt{\frac{g_m}{2R_4R_Lg_m+R_4+2R_L}}}$$

$$\begin{aligned} \text{K-LP: } & R_L \\ \text{K-HP: } & -\frac{R_4 R_L}{2R_4 R_L g_m + R_4 + 2R_L} \\ \text{K-BP: } & \frac{C_4 R_4 R_L g_m - C_5 R_L}{C_4 R_4 g_m + 2C_4 R_L g_m + 2C_5 R_L g_m + C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{\sqrt{-g_m}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}} \end{aligned}$$

$$10.6 \quad \mathbf{X-INVALID-WZ-6} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 R_5 R_L s^2 + R_5 R_L g_m - R_L + s (C_4 R_4 R_5 R_L g_m - C_4 R_4 R_L - C_5 R_5 R_L)}{R_5 g_m + 2R_L g_m + s^2 (2C_4 C_5 R_4 R_5 R_L g_m + C_4 C_5 R_4 R_5 + 2C_4 C_5 R_5 R_L) + s (C_4 R_4 R_5 g_m + 2C_4 R_4 R_L g_m + C_4 R_4 + 2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5) + 1}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5}R_Lg_m\sqrt{\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{2R_Lg_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{2R_Lg_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}+2\sqrt{C_4}\sqrt{C_5}\sqrt{R_5}R_L\sqrt{\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{2R_Lg_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}}{C_4R_4R_5g_m+2C_4R_4R_Lg_m+C_4R_4+2C_4R_5R_Lg_m+2C_4R_L+2C_5R_5R_Lg_m+C_5R_5}$$

$$\text{wo: } \sqrt{\frac{R_5g_m+2R_Lg_m+1}{2C_4C_5R_4R_5R_Lg_m+C_4C_5R_4R_5+2C_4C_5R_5R_L}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{R_5g_m+2R_Lg_m+1}{2C_4C_5R_4R_5R_Lg_m+C_4C_5R_4R_5+2C_4C_5R_5R_L}}(C_4R_4R_5g_m+2C_4R_4R_Lg_m+C_4R_4+2C_4R_5R_Lg_m+2C_4R_L+2C_5R_5R_Lg_m+C_5R_5)}{2\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5}R_Lg_m\sqrt{\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{2R_Lg_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{2R_Lg_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}+2\sqrt{C_4}\sqrt{C_5}\sqrt{R_5}R_L\sqrt{\frac{R_5g_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{2R_Lg_m}{2R_4R_Lg_m+R_4+2R_L}+\frac{1}{2R_4R_Lg_m+R_4+2R_L}}}$$

$$\begin{aligned} \text{K-LP: } & \frac{R_5 R_L g_m - R_L}{R_5 g_m + 2R_L g_m + 1} \\ \text{K-HP: } & -\frac{R_4 R_L}{2R_4 R_L g_m + R_4 + 2R_L} \\ \text{K-BP: } & \frac{C_4 R_4 R_5 R_L g_m - C_4 R_4 R_L - C_5 R_5 R_L}{C_4 R_4 R_5 g_m + 2C_4 R_4 R_L g_m + C_4 R_4 + 2C_4 R_5 R_L g_m + 2C_4 R_L + 2C_5 R_5 R_L g_m + C_5 R_5} \\ \text{Qz: } & \text{None} \end{aligned}$$

$$\text{Wz: } \frac{\sqrt{-R_5 g_m + 1}}{\sqrt{C_4} \sqrt{C_5} \sqrt{R_4} \sqrt{R_5}}$$

$$10.7 \quad \text{X-INVALID-WZ-7} \quad Z(s) = \left(\infty, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, R_L \right)$$

$$H(s) = \frac{R_L g_m + s^2 (C_4 C_5 R_4 R_5 R_L g_m - C_4 C_5 R_4 R_L) + s (C_4 R_4 R_L g_m + C_5 R_5 R_L g_m - C_5 R_L)}{g_m + s^2 (C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_4 R_L g_m + C_4 C_5 R_4 + 2 C_4 C_5 R_5 R_L g_m + 2 C_4 C_5 R_L) + s (C_4 R_4 g_m + 2 C_4 R_L g_m + C_5 R_5 g_m + 2 C_5 R_L g_m + C_5)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{C_4} \sqrt{C_5} R_4 R_5 g_m \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} + 2 \sqrt{C_4} \sqrt{C_5} R_4 R_L g_m \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} + \sqrt{C_4} \sqrt{C_5} R_4 \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} + 2 \sqrt{C_4} \sqrt{C_5} R_5 R_L g_m \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} + 2 \sqrt{C_4} \sqrt{C_5} R_L \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}}}{C_4 R_4 g_m + 2 C_4 R_L g_m + C_5 R_5 g_m + 2 C_5 R_L g_m + C_5}$$

$$\text{wo: } \sqrt{\frac{g_m}{C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_4 R_L g_m + C_4 C_5 R_4 + 2 C_4 C_5 R_5 R_L g_m + 2 C_4 C_5 R_L}}$$

$$\text{bandwidth: } \frac{\sqrt{C_4 C_5 R_4 R_5 g_m} \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} + 2 \sqrt{C_4} \sqrt{C_5} R_4 R_L g_m \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} + \sqrt{C_4} \sqrt{C_5} R_4 \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} (C_4 R_4 g_m + 2 C_4 R_L g_m + C_5 R_5 g_m + 2 C_5 R_L g_m + C_5)}{\sqrt{C_4} \sqrt{C_5} R_4 R_5 g_m \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} + 2 \sqrt{C_4} \sqrt{C_5} R_4 R_L g_m \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} + \sqrt{C_4} \sqrt{C_5} R_4 \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} + 2 \sqrt{C_4} \sqrt{C_5} R_5 R_L g_m \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}} + 2 \sqrt{C_4} \sqrt{C_5} R_L \sqrt{\frac{g_m}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}}}$$

K-LP: R_L

$$\text{K-HP: } \frac{R_4 R_5 R_L g_m - R_4 R_L}{R_4 R_5 g_m + 2 R_4 R_L g_m + R_4 + 2 R_5 R_L g_m + 2 R_L}$$

$$\text{K-BP: } \frac{C_4 R_4 R_L g_m + C_5 R_5 R_L g_m - C_5 R_L}{C_4 R_4 g_m + 2 C_4 R_L g_m + C_5 R_5 g_m + 2 C_5 R_L g_m + C_5}$$

Qz: None

$$\text{Wz: } \sqrt{\frac{g_m}{C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4}}$$

11 X-PolynomialError